

An Inspectable Design-to-Screen Agent for De Novo Protein Campaign Downselection

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Abstract

Modern generative protein-design systems can propose thousands of structurally plausible candidates, but experimental expression, binding, developability, and immune testing remain limiting steps. We present an inspectable scientific-agent workflow intended to reduce this design-to-experiment gap by combining literature-grounded design specifications, versioned de novo generation, structural triage, and route-aware immune-risk evidence. For MHC-I screening, overlapping 9-mers are represented by a frozen ESM-2 650M encoder, and a compact-student workflow is configured to imitate separate NetMHCpan 4.1 eluted-ligand (EL) and binding-affinity (BA) channels on Protein Design Archive sequences. Existing cleavage and TAP heads remain separate, and the processing composite is the geometric mean of N-terminal cleavage, C-terminal cleavage, and BA binding propensity; EL presentation and TAP are reported separately. MHC-II EL and BA predictions, CD4-response evidence, tolerance, and uncertainty are retained as distinct lanes; a combined rank is withheld when required evidence is unavailable. The completed full-PDA corpus contains 160,012 unique peptides with frozen 1,280-dimensional embeddings. Across five component-grouped held-out folds, EL and BA predictions reached mean Spearman correlations of 0.9431 and 0.9478 against teacher rank, respectively. On 159,038 unique PDA peptides scored against the same NetMHCpan EL teacher, the legacy Chao1 MHCflurry head achieved strong-binder average precision 0.060, while the out-of-fold student achieved 0.820. We also demonstrate provenance-preserving orchestration and sequence-to-structure visualization on one 99-residue de novo protein. These metrics measure teacher imitation. The system is not validated for immunogenicity prediction; its intended use is computational campaign downselection and experimental prioritization rather than estimation of a measured probability of human immunogenicity.

1 Introduction

Computational protein design has entered a high-throughput regime in which massively parallel generation and design-build-test-learn cycles can produce and evaluate large candidate libraries [15]. RFDiffusion enables target- and hotspot-conditioned backbone generation, while ProteinMPNN efficiently samples sequences compatible with designed backbones [18, 6]. The Protein Design

Archive provides a curated corpus of de novo designed protein sequences and structures [3]. This growing design capacity creates a practical asymmetry: candidate generation is increasingly inexpensive, while experimental characterization remains slow and resource intensive.

Immunogenicity is one source of late-stage attrition for therapeutic proteins. Uptake and processing by antigen-presenting cells can expose peptides on HLA class II molecules, activate CD4 T cells, and support anti-drug antibody responses that alter exposure, efficacy, or safety [19]. Computational epitope prediction can identify liabilities early, but presentation is only one step in a pathway that also depends on processing, T-cell recognition, tolerance, formulation, and patient context. Functional deimmunization studies therefore treat predicted immune risk and molecular function as competing objectives whose practical trade-offs require experimental measurement [13].

1.1 Machine-learning precedent

The computational strategy follows two established machine-learning patterns. Foundation models are pretrained on broad data and subsequently adapted to many downstream tasks [1]. Knowledge distillation trains a student against outputs from a more cumbersome teacher so that task-specific behavior can be reproduced in a more deployable model [8]. DistilBERT illustrates full model compression by reducing the teacher encoder itself [14]. Our implementation is different: NetMHCpan supplies pseudo-labels, frozen ESM-2 supplies peptide representations, and a compact multilayer head learns the task mapping. This is feature-based task distillation, not end-to-end compression of ESM-2.

The same transfer pattern appears across scientific foundation models. Protein language models learn reusable sequence representations; the Nucleotide Transformer adapts DNA representations to genomic prediction tasks; and scGPT adapts a generative model trained across millions of cells to several single-cell tasks [11, 5, 4]. These examples support the general technique of pairing broad self-supervised representations with small supervised adapters. They do not establish that the transferred representation or teacher remains calibrated on designed proteins, so distribution-shift and external-validation requirements remain part of our method.

Our premise is that an agent can narrow a large de novo campaign to a smaller, traceable set of candidates before expensive experiments. Paperclip converts literature into cited design and validation constraints; Proto executes RFDiffusion3, ProteinMPNN, and structure-prediction tools; and a LangGraph workflow records artifacts, enforces approval and reviewer gates, and renders pathway-specific evidence in a scientific workbench. The output is a prioritization dossier that helps scientists decide what to express and test next. It is not a substitute for expression, binding, MAPPs, T-cell, or anti-drug-antibody studies.

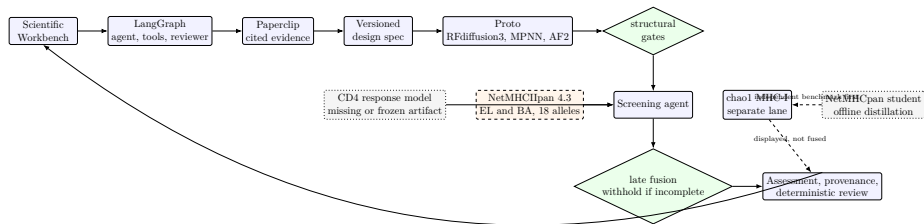


Figure 1: Inspectable design-to-screen architecture. Solid boxes are implemented in the LangGraph workbench path, dashed boxes use cached or frozen artifacts, and dotted boxes are offline or incomplete. Literature constrains generation rather than directly generating a valid protein. Structural gates precede route-aware immune evidence. The HLA-A*02:01 MHC-I lane remains separate from MHC-II late fusion, and a missing calibrated CD4 response artifact forces the combined rank to remain null.

2 Methods

2.1 Generation constraints

RFdiffusion established that target structures and interface hotspots can condition de novo binder generation [18]. We applied the newer all-atom RFdiffusion3 model through Proto on Modal [2], then used ProteinMPNN to sample sequences compatible with each generated backbone [6]. AlphaFold provides an independent refolding and interface-confidence check. Each target used a pinned PDB structure, literature- or complex-derived hotspots, two binder-length ranges (55–85 and 85–120 residues), and fixed low-temperature inference settings. The published in silico gates, monomer pLDDT above 80, interaction PAE below 10, and designed-to-refolded binder RMSD below 1 Å, are treated as triage thresholds rather than proof of binding.

2.2 Service architecture

The services have non-overlapping roles. Paperclip turns literature into cited design and validation constraints. Proto executes versioned biological tools and records their inputs, seeds, configurations, runtimes, and artifacts. IEDB provides live NetMHCIIpan EL and BA predictions. The local agent joins these outputs through typed adapters, records provenance, and renders sequence and structure-linked evidence. BenchFlow can package the validation lanes as separate frozen tasks with deterministic verifiers; it evaluates reproducibility and task completion, but does not create biological ground truth.

2.3 Frozen protein-language representations and processing heads

The MHC-I lane uses final-layer residue representations from `esm2_t33_650M_UR50D`, a 33-layer protein language model with 1,280-dimensional embeddings. Prior work supports transfer of ESM representations to peptide-MHC prediction, although published systems commonly fine-tune the encoder and include an MHC pseudo-sequence [7]. Our implementation instead freezes ESM-2 and trains compact task heads, reducing the number of trainable parameters, avoiding repeated teacher calls, and making the learned adapter easy to version. Inference still requires the 650M-parameter ESM-2 encoder, so the current system is not a fully compressed small model. A genuinely compact deployment would also distill or replace the encoder. Freezing ESM-2 creates an explicit domain-shift caveat because it was pretrained primarily on natural full-length proteins rather than designed proteins and isolated 9-mers.

The legacy `chao1` checkpoint supplies independently trained N- and C-terminal cleavage heads and a relative TAP transport head. It was associated with a 5,000,613-row HLA-A*02:01 corpus containing human, bacterial, viral, and de novo peptides, but most cleavage and MHC labels were teacher-derived and only 613 TAP measurements were experimental. The old MHCflurry-derived MHC head is therefore being replaced, while cleavage and TAP remain separately reported processing evidence. The checkpoint is an MHC-I surrogate, not an MHC-II, CD4-response, or human-immunogenicity model.

2.4 Full-PDA NetMHCpan distillation

The replacement student is trained on the full Protein Design Archive distribution rather than treating PDA as an external benchmark. Every overlapping 9-mer from every PDA parent is first labeled with NetMHCpan 4.1 EL and BA outputs. Exact peptide duplicates are then collapsed while retaining occurrence and parent counts. Parents sharing any exact 9-mer are connected into components, and each connected component is assigned wholly to one of five folds. Greedy fold balancing uses unique-peptide count, preventing direct peptide leakage while retaining all available designed-protein parents.

Each 9-mer is represented by the mean of its nine ESM-2 residue vectors, excluding special tokens. A two-output head applies layer normalization, a 256-unit hidden layer with GELU and dropout, and sigmoid EL and BA outputs. Teacher percentile rank r is transformed to a higher-is-riskier propensity

$$p(r) = 1 - \frac{\log(1+r)}{\log(101)}, \quad 0 \leq r \leq 100. \quad (1)$$

Five heads are trained. For each head, one component-grouped fold is untouched test data, the next fold is validation data, and the remaining three folds are training data. Deployment averages the five heads, while reported cross-validation metrics use only out-of-fold predictions. These metrics quantify NetMHCpan imitation, not biological accuracy.

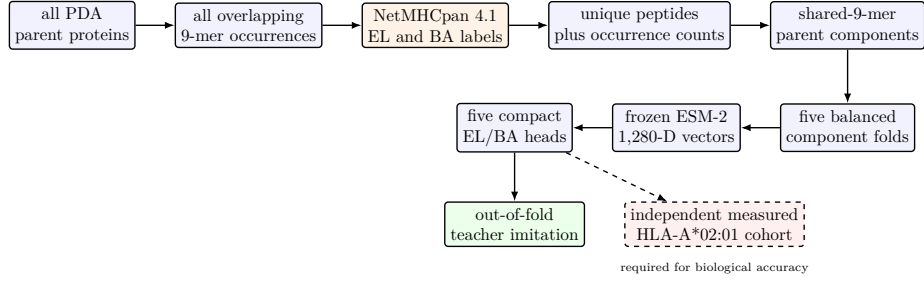


Figure 2: Full-PDA NetMHCpan student workflow. Exact shared 9-mers connect parent proteins before five-fold assignment, preventing direct peptide leakage across cross-validation folds. Held-out folds measure teacher imitation. Because the entire PDA corpus supplies training data across the ensemble, a separate non-PDA experimental cohort is required for out-of-distribution and biological-accuracy claims.

2.5 MHC-I risk calculation

For each 9-mer i , the student reports EL presentation propensity $p_{EL,i}$ and BA binding propensity $p_{BA,i}$. Predicted percentile ranks are deterministic inversions of the target transform. EL rank at or below 0.5 is labeled strong, rank above 0.5 and at or below 2.0 is labeled weak, and larger rank is labeled nonbinder. EL and BA are not averaged together because the channels are correlated and EL already includes presentation information. BA enters the current processing heuristic alongside cleavage, while EL and TAP remain separately reported evidence lanes.

The route-specific processing score combines the legacy cleavage heads with the replacement EL head:

$$R_i = \exp\left(\frac{\log \tilde{p}_{N,i} + \log \tilde{p}_{C,i} + \log \tilde{p}_{BA,i}}{3}\right), \quad \tilde{p} = \max(p, 10^{-6}). \quad (2)$$

For a protein with n windows, the displayed summary is the mean of the $K = \min(5, n)$ largest R_i values. TAP score, TAP uncertainty, EL propensity, the maximum window score, and residue-level projections are reported separately. This composite is a transparent prioritization heuristic, not a calibrated probability of immune response.

2.6 Immunogenicity constraints

For extracellular therapeutic binders, the primary biological axis is MHC-II presentation and CD4 T-cell response, not proteasomal MHC-I processing. Full candidate sequences are tiled into overlapping 15-mers. NetMHCIIpan 4.3 EL and BA are called separately through the IEDB Tools API across a predeclared 18-member HLA-DR, HLA-DP, and HLA-DQ panel. This separation follows the distinct semantics and limitations of eluted-ligand and binding-affinity evidence

[12]. TCR-facing self similarity can affect ranking only when the query and matched-self peptide share predicted HLA presentation. Presentation models can support early candidate filtering, but TCR recognition and downstream immune activation limit their direct predictive power for anti-drug antibodies [17].

2.7 Validation architecture

The orchestration layer accepts immutable model artifacts through versioned, capability-typed adapters. The validation pack keeps three lanes separate. First, `chao1` must be evaluated on independent HLA-A*02:01 eluted ligands with parent- and 9-mer-overlap exclusion, using MHC-only and cleavage-plus-MHC ablations. Second, NetMHCIIpan EL and BA are evaluated per allele on independent MHC-II presentation data, while a separate frozen IEDB class II T-cell benchmark evaluates response models. Therapeutic MAPPs families, including wild-type, immunogenic, and de-immunized factor VIIa variants, provide a compact biological ordering check [9]. Third, Protein Design Archive sequences supply the component-grouped five-fold distillation corpus [3]. PDA folds can measure held-out teacher imitation, but because all folds contribute to the deployed ensemble, a separate non-PDA cohort is required to measure out-of-distribution behavior or biological accuracy.

The joint result is a route-aware vector, not an average of incompatible scores. For an extracellular protein, MHC-II presentation and CD4 response are primary and the MHC-I lane is secondary evidence. For intracellular or nucleic-acid delivery, a high value in either validated pathway can trigger review. A predeclared late-fusion rule applies only within the calibrated MHC-II/CD4 lane; missing required evidence remains null and withholds the combined rank. Exact sequence hashes gate every residue-to-structure visualization.

3 Results

3.1 Design and orchestration

The reviewed RFdiffusion3 branch summary reports 100 backbone complexes for each of five targets, for 500 candidate backbones. The underlying backbone PDB files were not preserved in that snapshot, so the count demonstrates reported generation scale but is not yet auditable end to end. In a direct PD-L1 control check, 15 of the first 16 backbones contacted all three configured hotspots within 4.5 Å, and all 16 contacted all three within 6 Å. These are geometric backbone results only; sequence design, refolding, interface confidence, binding, and immune screening have not yet been completed on those exact generated backbones.

The workbench connects a LangGraph state machine to typed scientific tools, immutable artifact manifests, approval-gated compute, and deterministic reviewer checks. A live NetMHCIIpan control query completed both EL

Held-out metric	EL channel	BA channel
Cross-validation folds	5	5
Pooled out-of-fold peptides	160,012	160,012
Binder prevalence	0.0665 ± 0.0036	0.0341 ± 0.0026
Spearman correlation	0.9431 ± 0.0043	0.9478 ± 0.0043
Binder AUPRC	0.8914 ± 0.0090	0.8595 ± 0.0028
No-skill AUPRC	0.0665	0.0341
Propensity MAE	0.0557 ± 0.0020	0.0448 ± 0.0019
Propensity RMSE	0.0761 ± 0.0023	0.0660 ± 0.0022

Table 1: Cross-validated NetMHCpan teacher-imitation results. Values other than pooled peptide count are mean $\pm$ population standard deviation across five component-grouped held-out folds. Binder AUPRC uses the NetMHCpan rank threshold of 2 percent, and the no-skill reference equals binder prevalence. These are teacher-fidelity metrics, not validation of immunogenicity prediction.

and BA channels across all 18 predeclared HLA-II panel members. Because no calibrated CD4-response artifact was available, the late-fusion rule returned a null combined rank rather than substituting presentation evidence for response evidence.

3.2 MHC-I corpus and student results

The completed full-PDA corpus contains 1,602 parent proteins and approximately 290,937 overlapping 9-mer occurrences before exact-peptide deduplication. Deduplication produced 160,012 unique peptides with a frozen ESM-2 embedding matrix of shape $160,012 \times 1,280$. The earlier parent-batch smoke cache contained 52,901 EL-labeled windows, but the reported model results use the completed EL and BA corpus.

Across five untouched component-grouped test folds, mean EL Spearman correlation against NetMHCpan rank was 0.9431 with population standard deviation 0.0043; mean BA correlation was 0.9478 with standard deviation 0.0043. At the 2-percent binder threshold, mean area under the precision-recall curve was 0.8914 for EL and 0.8595 for BA. Mean propensity MAE was 0.0557 for EL and 0.0448 for BA. Pooled out-of-fold evaluation covered all 160,012 unique peptides. These results establish high teacher-imitation fidelity under exact-9-mer component grouping. They do not establish experimental presentation accuracy. A completed same-row comparison on 159,038 unique profile peptides found strong-binder average precision of 0.060 for the legacy Chao1 MHCflurry head and 0.820 for the out-of-fold student EL head. This comparison measures agreement with the NetMHCpan teacher, not experimental error.

3.3 Direct comparison with the legacy Chao1 head

Table 1 asks how faithfully the frozen-ESM-2 student reproduces NetMHCpan on held-out PDA peptide components. We also compared the legacy Chao1

MHC-I score	Spearman	Strong AUROC	Strong AP
Legacy Chao1 MHCflurry head	0.354	0.731	0.060
NetMHCpan student EL, out of fold	0.943	0.993	0.820

Table 2: Direct same-row comparison against NetMHCpan 4.1 EL on 159,038 unique PDA peptides. Strong binders use teacher EL rank at or below 0.5 percent. The student improves average precision by about 14-fold under 2.5-percent prevalence. Chao1 was calibrated to MHCflurry, so this is evidence for replacing the MHC head within this screening stack, not an experimental error rate or an immunogenicity endpoint.

MHCflurry-derived presentation head and the student EL head on the same 159,038 unique PDA 9-mers. NetMHCpan EL rank at or below 0.5 percent defined 3,956 strong binders. Because positives were only 2.5 percent of peptides, strong-binder average precision was the primary discrimination metric; AUROC is reported as a secondary measure.

The same-row result justifies replacing the legacy Chao1 MHC head, while retaining its separately supervised cleavage and TAP heads. It does not show that the student exceeds NetMHCpan, because NetMHCpan is the teacher. Independent measured presentation and activation cohorts remain necessary for biological validation.

3.4 De novo protein demonstration

We used `pda:9s14:0`, a 99-residue PDA design in the “novel” structural bin with nearest-natural Foldseek TM-score 0.4728, as a traceable visualization example. The legacy `chao1` adapter scored all 91 overlapping 9-mers. Its top-five mean processing composite was 0.6975, and its maximum was 0.7053 at peptide `WVLATEIYR`. The parent sequence and all 91 9-mers were absent from the supplied legacy 5,000,613-row teammate parquet. NetMHCIIpan EL and BA also completed successfully, while the combined MHC-II/CD4 rank remained null because the response adapter was unavailable.

This demonstration establishes that a complete de novo sequence can be tiled, embedded, scored, mapped back to structure, and reviewed with provenance. It does not establish generalization of the new PDA-trained student. Once the full PDA corpus is used for training, PDA examples including 9S14 cannot serve as external OOD validation unless their complete connected component is excluded from every deployed head.

3.5 Neo-2/15 qualitative in vivo case

Neo-2/15 is an experimentally optimized de novo IL-2/IL-15 mimetic with a published 100-residue designed core (Supplementary Table S4 design `G2_neo2_40_1F_seq36_S11`) and an apo crystal structure (PDB 6DG6) [16]. The crystallographic polymer contains an additional N-terminal GSHM cleavage scar. The Methods also report

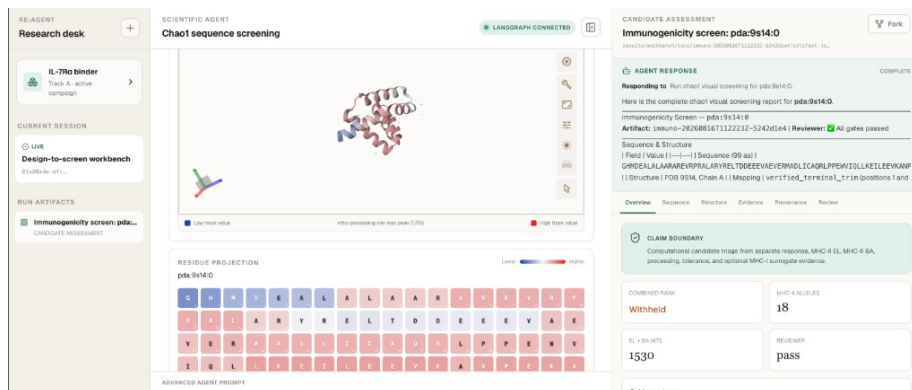


Figure 3: Scientific Workbench view of the 9S14 de novo protein demonstration. Residue-level processing tracks are linked to the three-dimensional structure and shown beside separate MHC-I, MHC-II, provenance, and reviewer evidence. The reviewer pass verifies artifact completeness and claim boundaries, not biological accuracy. The displayed processing score is a legacy-model smoke inference, and the combined MHC-II/CD4 rank is withheld.

Sequence variant	Length	Chao1 top five	Student top five	Strong	Weak
Designed core	100	0.732	0.594	3	7
PDB 6DGG chain	104	0.732	0.594	3	7
His-tagged expression construct	121	0.732	0.594	3	7

Table 3: Direct qualitative Neo-2/15 sensitivity analysis. Chao1 top-five scores combine its N-cleavage, C-cleavage, and legacy MHCfurry-derived presentation propensity. Student-hybrid top-five scores use the same cleavage heads with NetMHCpan-student BA propensity. Strong and weak counts apply only to the student EL-rank thresholds of 0.5 and 2.0 percent. The two composite scales are not calibrated as equivalent immune-response probabilities.

a 21-residue N-terminal His-tag and thrombin-site construct; because the exact construct administered in every animal dose is not stated unambiguously, we scored the designed core, the 104-residue crystallographic chain, and the 121-residue tagged construct as a sensitivity analysis.

For the designed core, Chao1 assigned a top-five mean composite of 0.732 and maximized at YDALMILNI with 0.759. The student hybrid assigned a top-five mean of 0.594 and maximized at the overlapping DALMILNIV window with 0.642. Both techniques therefore localized their highest composite processing evidence to the same central core region, while differing in ordering and magnitude. Separately, the student classified ALYDALMIL, YAFNFELIL, and AIITILQSW as strong HLA-A*02:01 presentation candidates, with predicted EL ranks of 0.035, 0.047, and 0.397 percent, respectively. Neither construct extension changed the three strong and seven weak student calls or either model’s top-five mean.

Case	Empirical system			HLA-A*02:01 alignment	Model summary	Reported outcome
Neo-2/15 core	C57BL/6 serum IgG	in vivo	None		3 strong, 6 weak; top-five 0.584	No detectable anti-protein IgG
Latent-X2 TNFL9 VHH	Human PBMC proliferation cytokines		3 of 10 donors		1 strong, 7 weak; top-five 0.566	No detectable proliferation or cytokine elevation

Table 4: Qualitative comparison of sequence-resolved de novo protein cases. Counts and scores are model outputs; empirical systems and endpoints differ and are not accuracy labels for the MHC-I model.

Silva and colleagues administered Neo-2/15 to C57BL/6 mice by daily intraperitoneal injection at 10 micrograms per day for 14 days (10 animals per group), then assayed serum IgG by ELISA; they reported no detectable anti-Neo-2/15 response [16]. The apparent discordance is scientifically expected and must not be presented as human immunogenicity validation. Both profiles represent model-dependent human HLA-A*02:01 MHC-I processing hypotheses, whereas the experiment measured whole-animal murine IgG and depended on murine MHC, MHC-II and CD4 help, B-cell epitopes, tolerance, dose, formulation, and assay sensitivity. This case therefore demonstrates a useful liability-triage output and its limitation: predicted presentation candidates can motivate follow-up experiments, but they neither predict nor explain the reported mouse ADA result.

3.6 Human donor case: a de novo TNFL9 VHH

A closer species-aligned case is provided by `LX2_VHH_TNFL9_10`, a 128-residue de novo-designed VHH domain generated with Latent-X2. Its complete domain sequence and 31.1 nM measured TNFL9 affinity are published [10]. This candidate was one of four TNFL9 VHHs tested ex vivo against peripheral-blood mononuclear cells from ten healthy human donors. Three donors carried HLA-A*02:01, including one homozygous donor. Across antibody concentrations up to 30 micrograms per milliliter, the study reported no increase in T-cell proliferation at 48 or 120 hours and no cytokine elevation at 120 hours relative to background [10].

The hybrid model scored all 120 overlapping 9-mers in the published VHH domain. It classified one window as strong and seven as weak by predicted NetMHCpan EL rank. The strong window, `MVYLMNSL` at VHH positions 78–86, had predicted EL rank 0.235 percent, BA rank 0.172 percent, and composite processing score 0.686. The protein-level top-five processing mean was 0.566. The strongest window lies in a conserved framework-like segment rather than the designed CDR3 tail, illustrating why presentation propensity and immune novelty are not equivalent.

This human donor case removes the murine-species mismatch but does not constitute biological validation. The assayed antibodies were human IgG1 Fc

fusions, while only the VHH domain sequence was published and scored. The PBMC assay was not peptide-resolved or CD8-specific, donor-level outcomes were not reported by HLA genotype, and it measured neither MHC-I presentation nor clinical anti-drug antibodies. The valid conclusion is narrower: a predicted strong HLA-A*02:01 processing liability can coexist with no detectable response in a small human ex vivo panel. This supports using the model to locate liabilities for follow-up, while demonstrating that MHC-I presentation scores cannot be interpreted as calibrated human immunogenicity probabilities.

4 Discussion

4.1 From generative scale to an experimental shortlist

The intended value is not another unconstrained generator. It is a decision layer between rapid candidate generation and expensive experiments. Computational design campaigns can begin with tens of thousands of models and reduce them to hundreds or fewer for testing; empirical design-build-test-learn feedback remains decisive even for structurally plausible candidates [15]. Adding route-aware immune evidence at this stage allows scientists to compare candidates that satisfy structural gates, identify localized peptide liabilities, select naturally lower-risk alternatives, or propose constrained redesigns before expression and animal work.

As de novo protein generation becomes increasingly ubiquitous, the practical bottleneck shifts from proposing candidates to deciding which candidates justify scarce experimental capacity. Our mission is to make that transition easier by providing a modular, provenance-aware filtering layer that can plug into existing generation, structure, and design-build-test-learn pipelines. The system is designed to help teams downselect at campaign scale, shorten the path from in silico proposals to in vivo testing, and preserve a clear record of why candidates advanced. It does not replace experimental immunogenicity assessment or claim that a computational score predicts a human immune response.

4.2 Applicability to de novo proteins

MHC molecules ultimately encounter peptide sequences, so every canonical de novo protein can be tiled and represented in the same peptide space as a natural protein. Training the compact head on embeddings from PDA-derived peptides adapts the downstream model to the designed-protein distribution. Nevertheless, both the frozen ESM-2 representation and the NetMHCpan teacher were developed primarily from natural sequence data. The resulting score should therefore be treated as a transfer-learning hypothesis under distribution shift. A newly generated non-PDA candidate with no exact or near-neighbor overlap to training is an appropriate demonstration case; only independent measured eluted-ligand or binding data can establish biological accuracy.

4.3 Experimental handoff

The proposed handoff is a ranked, diverse panel rather than a single winner. Structural gates should be followed by expression and solubility assays, target-binding measurements, refolding or structural confirmation, HLA-II MAPPs, donor-diverse PBMC or T-cell assays, and anti-drug-antibody evaluation when appropriate. Computational deimmunization studies show that predicted epitope reduction and retained molecular function form a Pareto frontier, and that experimental measurements are required to quantify the trade-off [13]. The agent records why each candidate was selected and which evidence remains missing so negative experimental results can improve the next design round.

5 Intended use and claim boundary

The system has not been validated for immunogenicity prediction. It is intended as a scalable filtering and prioritization layer that reduces time and manual effort between a therapeutic design hypothesis and a traceable set of sequences prioritized for experimental testing. Backbone contact, structure-prediction confidence, HLA presentation, processing surrogates, and response-model scores are distinct evidence types. They are not presented as clinical predictions, measured T-cell responses, proof of binding, or experimental validation. Wet-lab expression, binding, presentation, T-cell, and anti-drug-antibody studies remain the decision authority.

6 Limitations and next experiments

1. **Generation: reported, not yet auditable end to end.** Recover or rerun a small backbone set, preserve the PDBs and hashes, then perform ProteinMPNN sequence design, refolding, and interface QC.
2. **Literature integration: demonstrated.** Literature determined the hotspot-conditioned generator, structural gates, MHC-II biological axis, 15-mer unit, and separate EL/BA channels. The final demo should show these citations beside the corresponding configuration.
3. **Constraint design: partially demonstrated.** PD-L1 hotspot contact is measured, but target-specific clash, refold RMSD, monomer confidence, and interface PAE gates remain to be run on sequence-bearing candidates.
4. **Immunogenicity prediction: not validated.** The MHC-I student currently measures NetMHCpan teacher imitation, and the MHC-II screen reports presentation evidence while missing response evidence is surfaced. Independent measured presentation or binding cohorts, a calibrated response artifact, donor-diverse T-cell assays, and anti-drug-antibody studies are required before any biological or clinical immunogenicity claim. Until then, these outputs are campaign-scale downselection filters only.

5. **Creative sponsor-tool use: demonstrated in components.** Proto generated the backbones, Paperclip grounded the constraints, IEDB supplied live HLA predictions, and the workbench preserves provenance and sequence-verified 3D overlays. BenchFlow is a proposed standalone packaging layer for the frozen validation tasks. The final demo must connect the implemented components in one candidate trace.

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